Finding the best bundle

Lecture notes

What this lecture is about. Last lecture ended with a picture: the best feasible bundle is where the highest reachable indifference curve touches the budget line, and there $MRS = p_x/p_y$. Today we find that bundle with numbers, in four ways that all say the same thing: the picture, "bang for the buck", arrows (vectors), and the Lagrangian. Then we practise, and we look at the cases where the rule does not apply: corners and kinks.

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1 One example for the whole lecture

Example: the running example (the same as at the end of lecture 8)

Utility $u = x\,y$; prices $p_x = 3$, $p_y = 1$; money $B = 60$. The budget line is $3x + y = 60$. Three bundles on it:

$$A = (5, 45), \qquad C = (15, 15), \qquad E = (10, 30).$$

Check: $3 \cdot 5 + 45 = 3 \cdot 15 + 15 = 3 \cdot 10 + 30 = 60$. Which one is the best feasible bundle?

2 Way 1: the picture

A and C are equally good: $u(A) = 5 \cdot 45 = 225 = 15 \cdot 15 = u(C)$, so they lie on the same indifference curve, $u = 225$. That curve *cuts* the budget line. At A the curve is steeper than the line, so moving along the line toward more x reaches better curves; at C it is flatter, so moving back toward more y helps. At E the curve $u = 10 \cdot 30 = 300$ just touches the line: the same slope, and no better curve is reachable. E is the best feasible bundle.

3 Way 2: bang for the buck

Marginal utility

MU_x is how much u rises with one more unit of x, holding y fixed (a derivative). For $u = xy$: $MU_x = y$ and $MU_y = x$. Check at A: from $(5, 45)$ to $(6, 45)$, u goes from 225 to 270, up by $45 = y$.

Divide each marginal utility by its price: utility per euro.

bundle	MU_x/p_x	MU_y/p_y	a euro moved ...
$A = (5, 45)$	$45/3 = 15$	$5/1 = 5$	from y to x gains
$C = (15, 15)$	$15/3 = 5$	$15/1 = 15$	from x to y gains
$E = (10, 30)$	$30/3 = 10$	$10/1 = 10$	gains nothing

At the best bundle, the last euro does equally well on both goods. Keep the number 10 in mind.

4 Way 3: arrows

- **Along the line.** One more x and three less y cost the same: the step $d = (1, -3) = (p_y, -p_x)$ keeps you on the budget line, and so does any multiple of it, big or small.
- **The prices.** The price vector $p = (3, 1)$ is perpendicular to the line: $p \cdot d = 3 \cdot 1 + 1 \cdot (-3) = 0$.
- **Better.** The arrow of marginal utilities points toward better bundles, across the indifference curve.

Gradient

The gradient $\nabla u = (MU_x, MU_y)$ is the arrow made of the two marginal utilities. It points in the direction in which utility rises fastest, and it is perpendicular to the indifference curve.

Project the gradient onto the line. Multiply it with the step d:

bundle	∇u	$\nabla u \cdot d$	along the line
$A = (5, 45)$	$(45, 5)$	$45 - 15 = 30$	toward more x is better
$C = (15, 15)$	$(15, 15)$	$15 - 45 = -30$	toward more y is better
$E = (10, 30)$	$(30, 10)$	$30 - 30 = 0$	no step helps

At E the gradient has nothing left along the line: it is perpendicular to the line, like p. Two arrows perpendicular to the same line are multiples of each other (both point up and to the right, so the multiple is positive):

$$\nabla u = \lambda\, p, \qquad (30, 10) = 10 \cdot (3, 1).$$

Insight: three pictures, one condition

$\nabla u \cdot d = MU_x\, p_y - MU_y\, p_x = 0$ is the same as $\dfrac{MU_x}{MU_y} = \dfrac{p_x}{p_y}$ (MRS = price ratio, way 1) and as $\dfrac{MU_x}{p_x} = \dfrac{MU_y}{p_y}$ (equal bang for the buck, way 2).

5 Way 4: the Lagrangian

Lagrangian

$\mathcal{L} = u(x,y) + \lambda\,(B - p_x x - p_y y)$: utility, plus a number λ times the money left over. Setting its derivatives to zero gives the arrow condition and the budget line at once.

For our example, $\mathcal{L} = xy + \lambda\,(60 - 3x - y)$, and

$$\frac{\partial \mathcal{L}}{\partial x} = y - 3\lambda = 0, \qquad \frac{\partial \mathcal{L}}{\partial y} = x - \lambda = 0, \qquad \frac{\partial \mathcal{L}}{\partial \lambda} = 60 - 3x - y = 0.$$

The first two say $(y, x) = \lambda\,(3, 1)$: the gradient is a multiple of the price vector (way 3). The third is the budget line. Solve: $x = \lambda$, $y = 3\lambda$, so $y = 3x$ (MRS $y/x = 3$ = price ratio); then $3x + 3x = 60$ gives $x = 10$, $y = 30$, and $\lambda = 10$, the bang for the buck at E.

The Lagrange multiplier λ

At the best bundle, λ measures how much utility one more euro of money buys.

With money B the best bundle is $x = B/6$, $y = B/2$, so $u^* = B^2/12$. Give more money in smaller and smaller steps:

B	best bundle	u^*	$\Delta u^*/\Delta B$
60	$(10, 30)$	300	
61	$(10.167, 30.5)$	310.083	10.083
60.1	$(10.017, 30.05)$	301.0008	10.008
60.01	$(10.0017, 30.005)$	300.1000	10.0008

The ratio settles at $\lambda = 10$. (The number depends on how we number utility: with $u = 2xy$, λ doubles to 20. Its meaning, utility per extra euro at the best bundle, does not.)

6 Worked exercises

Example: 1 · climbing and cinema

You spend $B = 72$ euros a week on climbing visits (x, €6) and cinema tickets (y, €12); $u = x^{1/3}y^{2/3}$.

- $MU_x = \frac{1}{3}x^{-2/3}y^{2/3}$, $MU_y = \frac{2}{3}x^{1/3}y^{-1/3}$, so $MRS = \dfrac{y}{2x}$.
- $\dfrac{y}{2x} = \dfrac{6}{12}$ gives $y = x$; the budget $6x + 12y = 72$ gives $(4, 4)$, with $u^* = 4$.
- Bang for the buck at $(4,4)$: $\dfrac{1/3}{6} = \dfrac{2/3}{12} = \dfrac{1}{18}$. The Lagrangian gives the same bundle

and $\lambda = \dfrac{1}{18}$.

- For any B, $x = y = B/18$ and $u^* = B/18$: here every extra euro adds exactly $\lambda = 1/18$.

Example: 2 · gel pens and pencils (a corner)

$u = 3x + 2y$ (x gel pens, y pencils), so $MRS = 3/2$ everywhere; budget €12, a pencil €1. With a gel pen at €2 the price ratio is $2 > 3/2$: all on pencils, $(0, 12)$. At €1.20 the ratio is $1.2 < 3/2$: all on gel pens, $(10, 0)$.

Example: 3 · toy tricycles (a kink)

Each tricycle needs 1 frame and 3 wheels: $u = \min\{3x, y\}$. A frame costs €30, a wheel €10, budget €180. One tricycle costs €60, so the best bundle is $(3, 9)$. At the kink there is no single slope, so "MRS = price ratio" cannot be used; anything off the ratio $y = 3x$ is wasted money.

7 When the rule does not apply

The rule finds a best bundle where an indifference curve *touches* the line away from its ends. It works because the curves bow in: on a curve that bowed out, the touching point would be the worst bundle on the line. Two exceptions:

- **Corners.** If a step along the line always helps in one direction, you keep going until the line ends: all money on one good. There the gradient is not a multiple of p, yet the corner is the best feasible bundle, because the only improving step points off the end of the line.
- **Kinks.** If the indifference curve has a corner, its slope is not defined there.

Perfect substitutes. $u = 2x + 3y$, so $MRS = 2/3$ everywhere; $p_y = 3$, $B = 12$. With $p_x = 1$ the price ratio is $1/3 < 2/3$: all on x, $(12, 0)$. With $p_x = 3$ it is $1 > 2/3$: all on y, $(0, 4)$. With $p_x = 2$ it equals the MRS, and every point on the line is best. In arrows: $\nabla u = (2, 3)$ is a multiple of $p = (p_x, 3)$ only when $p_x = 2$.

Perfect complements. $u = \min\{x, y\}$, $p_x = 1$, $p_y = 3$, $B = 20$: buy in the ratio $x = y$, so $x + 3x = 20$ and the best bundle is $(5, 5)$.

Quasi-linear. $u = 2x + \sqrt{y}$, so $MU_x = 2$, $MU_y = 1/(2\sqrt{y})$ and $MRS = 4\sqrt{y}$; take $p_x = 8$, $p_y = 1$. In the interior $4\sqrt{y} = 8$ gives $y = 4$, whatever the money. With $B = 20$: $x = (20 - 4)/8 = 2$, the bundle $(2, 4)$. With $B = 2$ there is not even enough for $y = 4$: all on y, the corner $(0, 2)$. Once $y = 4$ is reached, extra money goes only into x.

Summary

- Inside the budget line, the best feasible bundle satisfies $MRS = p_x/p_y$; equivalently equal bang for the buck, or $\nabla u = \lambda p$.
- The Lagrangian turns this into algebra; λ is the utility one more euro buys at the best bundle.
- Corners: compare the MRS with the price ratio and spend everything on one good. Kinks: buy in the fixed ratio.

Further reading. Frank, *Microeconomics and Behavior*, pp. 71–76 and Appendix 3A, pp. 90–92 (week 5 page, additional readings; the password is given in class).